SwiftShip Tracker

Parcel booking and tracking on Salesforce with Agentforce, Prompt Builder and Flow

Project Report

Project	SwiftShip Tracker (Salesforce, Agentforce and Prompt Builder)
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Team Members	Saranraj, Somesh, Harristarun, Sriram Kaviyan
Platform	Salesforce Developer Edition
GitHub	github.com/Git-it-Bharathy

1. Introduction

1.1 Project Overview

SwiftShip Tracker is a Salesforce CRM solution that simplifies parcel booking, shipment tracking and delivery management. It stores parcel, delivery, sender and receiver information in custom objects with relationships between them. An Agentforce AI agent, built with Prompt Builder and Salesforce Flow, lets a user retrieve parcel details conversationally by giving a Parcel ID.

1.2 Purpose

- Centralize parcel and delivery information
- Simplify parcel booking and tracking
- Provide real-time shipment visibility
- Enable conversational parcel tracking using Agentforce and Prompt Builder
- Provide role-based access using permission sets

2. Ideation Phase

2.1 Problem Statement

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	Customer / Sender	book a parcel easily	booking requires support assistance or multiple applications	parcel operations are not centralized	frustrated
PS-2	Customer / Sender	track my parcel	I do not have a simple real-time tracking experience	shipment information is difficult to access	confused
PS-3	Delivery Agent	update parcel status	status updates require manual handling	there is a need for centralized operational tracking	concerned
PS-4	Customer	receive delivery updates	notifications may not be available at every delivery stage	communication needs to be automated	uncertain
PS-5	Admin	monitor parcel operations	parcel flow and delivery performance are difficult to monitor	operational data needs centralized reporting	concerned

2.2 Empathy Map Canvas

User persona: Customer / Sender

- **Think & Feel:** Wants a simple way to book and track parcels without repeatedly contacting support.
- **Hear:** Delivery confirmations and status updates from the parcel service.
- **See:** Parcel details, tracking status, estimated delivery date and delivery information.
- **Say & Do:** Books parcels, checks tracking information and asks for delivery updates.
- **Pain:** Delays, confusion, lack of easy tracking and dependence on support or multiple apps.
- **Gain:** Real-time visibility, self-service tracking, convenient booking and timely notifications.

2.3 Brainstorming and Idea Prioritization

Ideas considered: manual registers, spreadsheet tracking, a standalone parcel portal, and Salesforce CRM with Agentforce AI.

Final selection: Salesforce CRM with Agentforce AI, because it offers centralized parcel management, automation through Flows, real-time tracking visibility, conversational AI assistance and secure role-based access.

3. Requirement Analysis Phase

3.1 Customer Journey Map

Parcel inquiry, parcel booking, parcel record creation, dispatch, in-transit tracking, out for delivery, delivery confirmation, customer notification and reporting.

3.2 Solution Requirements

Functional requirements

FR No	Requirement	Status in this project
FR-1	Parcel booking and record creation	Implemented (Parcel object and records)
FR-2	Parcel status tracking	Implemented (Status picklist)
FR-3	Weight management	Implemented (Weight field)
FR-4	Estimated delivery date management	Implemented (Estimated Delivery Date field)
FR-5	Sender information management	Implemented (Sender object)
FR-6	Receiver information management	Implemented (Receiver object)
FR-7	Delivery and current-location tracking	Implemented (Delivery object, geolocation field)
FR-8	Agent parcel status updates	Planned (future scope)
FR-9	Automated customer notifications	Planned (future scope)
FR-10	AI-powered parcel tracking	Implemented (Agentforce agent, Prompt Builder, Flow)
FR-11	Reports and dashboards	Planned (future scope)
FR-12	Customer access through Experience Cloud	Planned (future scope)

Non-functional requirements

NFR	Description
Usability	Intuitive Salesforce interface and conversational agent
Security	Permission set controlling object and field access
Reliability	Accurate parcel status and delivery information
Performance	Agent responds within a few seconds (the test trace shows about 4 seconds for the Flow action)
Availability	Cloud-based Salesforce access
Scalability	Supports growing parcel and delivery data
Data integrity	Relationship-based parcel data
Maintainability	Flow, prompt and agent configuration can be updated

3.3 Data Flow

Customer / Sender, parcel booking, parcel record, sender and receiver information, delivery record, current location and estimated delivery, customer tracking through Agentforce.

AI tracking flow: customer query, Agentforce agent, subagent (Parcel Details Inquiry), Flow (Parcel Details), prompt template (Retrieve Parcel Details), parcel record, then a conversational response with Parcel ID, Status, Weight and Estimated Delivery Date.

3.4 Technology Stack

Layer	Technology
Platform	Salesforce Developer Edition
Database	Salesforce custom objects
Business logic and automation	Autolaunched Flow
AI	Agentforce (Agent Builder), Prompt Builder
Security	Permission set

4. Project Design Phase

4.1 Problem-Solution Fit

Problem	Solution	Benefit
Parcel booking is difficult	Salesforce parcel management	Centralized booking
Tracking requires support	Agentforce AI and Prompt Builder	Self-service tracking
Manual status updates	Salesforce Flows	Automated processes
Delivery communication gaps	Email Alerts (future scope)	Timely notifications
Scattered shipment information	Parcel, Delivery, Sender and Receiver objects	Structured data
Difficult operational monitoring	Reports and dashboards (future scope)	Better visibility
Different users need different access	Permission sets	Secure access

4.2 Proposed Solution

Parameter	Description
Problem	Difficult and inefficient parcel booking and tracking
Solution	Salesforce-based parcel management system
Innovation	Agentforce AI with Prompt Builder for conversational self-service tracking
Impact	Centralized data, real-time tracking visibility and less manual effort
Scalability	Can grow to Experience Cloud and external logistics integration

4.3 Solution Architecture

Users (customer, agent, admin) use Salesforce Lightning, which sits on the SwiftShip Tracker app. The data layer holds Parcel, Delivery, Sender and Receiver. The automation layer is the Parcel Details Flow. The AI layer is the Agentforce agent with the Prompt Builder template and the Parcel Details Inquiry subagent. The security layer is the Swift Ship permission set.

5. Project Planning and Scheduling

5.1 Product Backlog and Sprint Schedule

Sprint	Epic	USN	User story / task	Points	Priority
Sprint-1	Developer setup	USN-1	As a system administrator, I want to create and configure the Salesforce environment so that I can build the parcel management system.	3	High
Sprint-2	Data modeling	USN-2	As an admin, I want to create Parcel, Delivery, Sender and Receiver objects with relationships so that parcel data is stored systematically.	5	High
Sprint-2	Data modeling	USN-3	As a staff member, I want tabs for each module so that I can navigate easily.	5	High
Sprint-3	Automation	USN-4	As a system, I want to validate parcel details so that incomplete records are prevented.	3	High
Sprint-3	Automation	USN-5	As a delivery coordinator, I want a Flow to retrieve parcel information so that processing requires minimal manual effort.	3	High
Sprint-4	Apex	USN-6	As a system, I want Apex triggers and classes for critical parcel actions.	5	High
Sprint-4	Apex	USN-7	As an operations manager, I want Batch Apex jobs to keep records up to date.	5	High
Sprint-5	Reports	USN-8	As a manager, I want reports on parcel status and delivery performance.	4	High
Sprint-6	Dashboards and AI	USN-9	As a manager, I want dashboards, while Agentforce provides AI-powered tracking assistance.	4	Medium

5.2 Sprint Tracker

Sprint	Total story points	Completed	Status
Sprint-1	3	3	Completed
Sprint-2	10	10	Completed
Sprint-3	6	3	Partly completed (USN-5 done, USN-4 future scope)
Sprint-4	10	0	Future scope
Sprint-5	4	0	Future scope
Sprint-6	4	2	Partly completed (Agentforce done, dashboards future scope)

Stories USN-1, USN-2, USN-3 and USN-5, and the Agentforce part of USN-9, are implemented. USN-4, USN-6, USN-7, USN-8 and the dashboard part of USN-9 are future scope.

6. Project Development Phase

Phase 1: Requirement Analysis and Planning

Objective: Customers face delays and confusion while booking and tracking parcels because they need to call support or use several apps. A conversational AI agent can simplify this.

Work done:

- Identified users: customer/sender, delivery agent, support staff and admin
- Defined the scope: parcel booking, tracking and delivery information in one Salesforce org
- Designed the data model: Parcel, Delivery, Sender and Receiver with lookup relationships
- Selected the tools: Salesforce objects, Prompt Builder, Flow and Agentforce

Phase 2: Salesforce Development (Backend and Configuration)

Milestone 1: Salesforce account

Created a Developer Edition org at developer.salesforce.com/signup and verified the account.

Milestone 2: Objects

Created four custom objects (Parcel, Delivery, Sender, Receiver) with Allow Reports, Allow Search and Track Field History enabled.

Milestone 3: Tabs

Created a custom tab for each object.

Milestone 4: Fields and relationships

Object	Field	Type
Parcel	Parcel ID	Auto number P-{000}
Parcel	Status	Picklist (Booked, In Transit, Out for Delivery, Delivered)
Parcel	Weight	Number
Parcel	Estimated Delivery Date	Date
Parcel	Sender	Lookup (Sender)
Delivery	Current Location	Geolocation
Delivery	Estimated Delivery Date	Date
Delivery	Sender, Parcel	Lookup
Sender	Sender Address	Geolocation
Sender	Sender Contact	Phone
Sender	Sender Email	Email
Receiver	Receiver Address	Geolocation
Receiver	Receiver Contact	Phone
Receiver	Receiver Email	Email
Receiver	Sender, Parcel	Lookup

Sample data

Created a Sender (Rahul), a Parcel (Laptop Parcel, P-001, Booked, 10 kg), a Receiver (Priya) and a Delivery record, linked together.

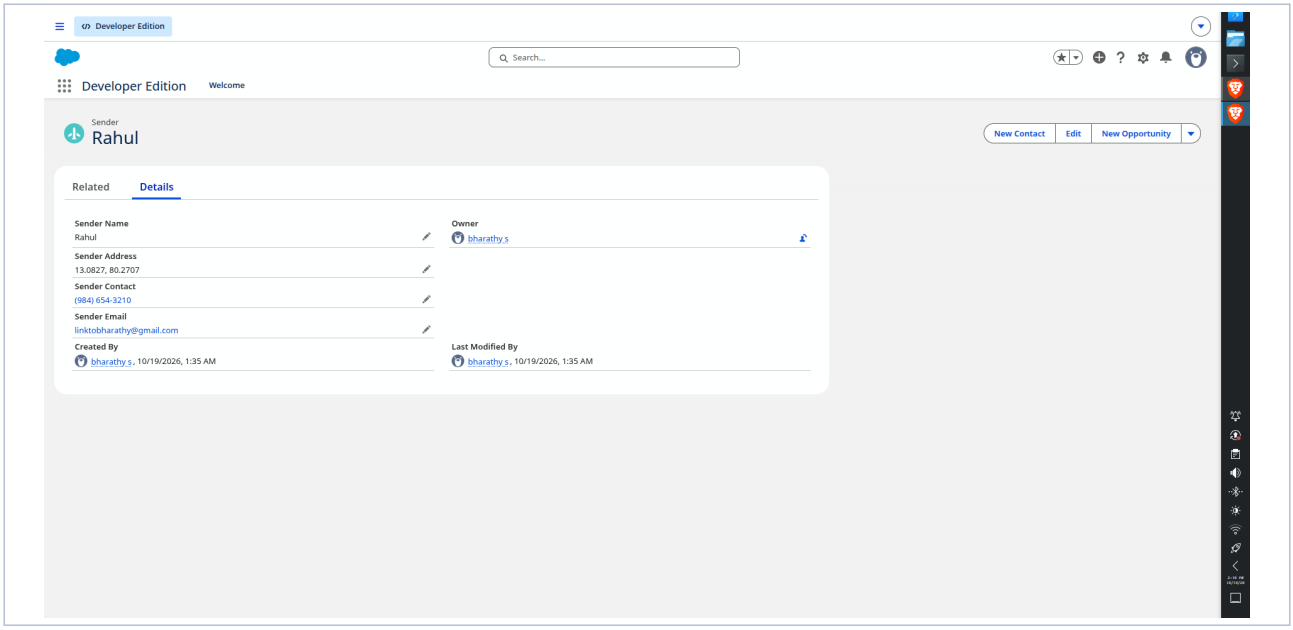


Figure 1: Sender record (Rahul)

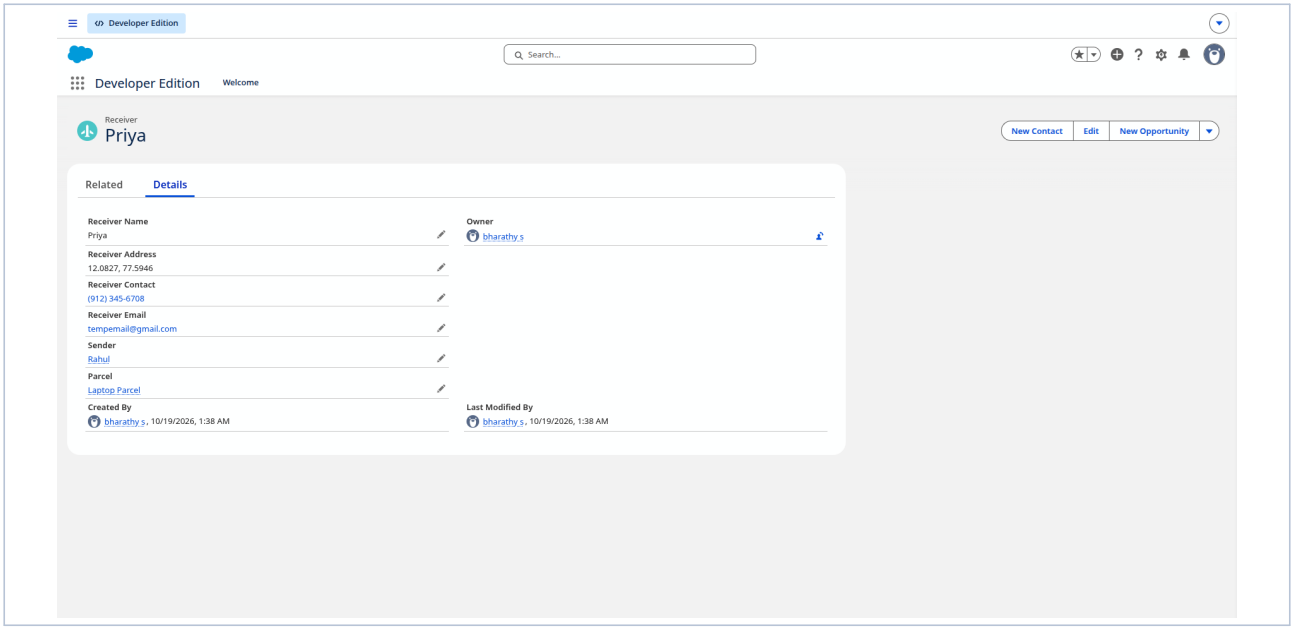


Figure 2: Receiver record (Priya) linked to Sender Rahul and Parcel Laptop Parcel

Milestone 5: Prompt Builder

Created a Record Summary prompt template, **Retrieve Parcel Details** (API name Retrieve_Parcel_Details), on the Parcel object. It returns the Parcel Name, Parcel ID, Status, Weight and Estimated Delivery Date. The prompt wording was refined after the first preview so that the model uses the supplied parcel details directly. The final preview returns the correct parcel update and the template is active.

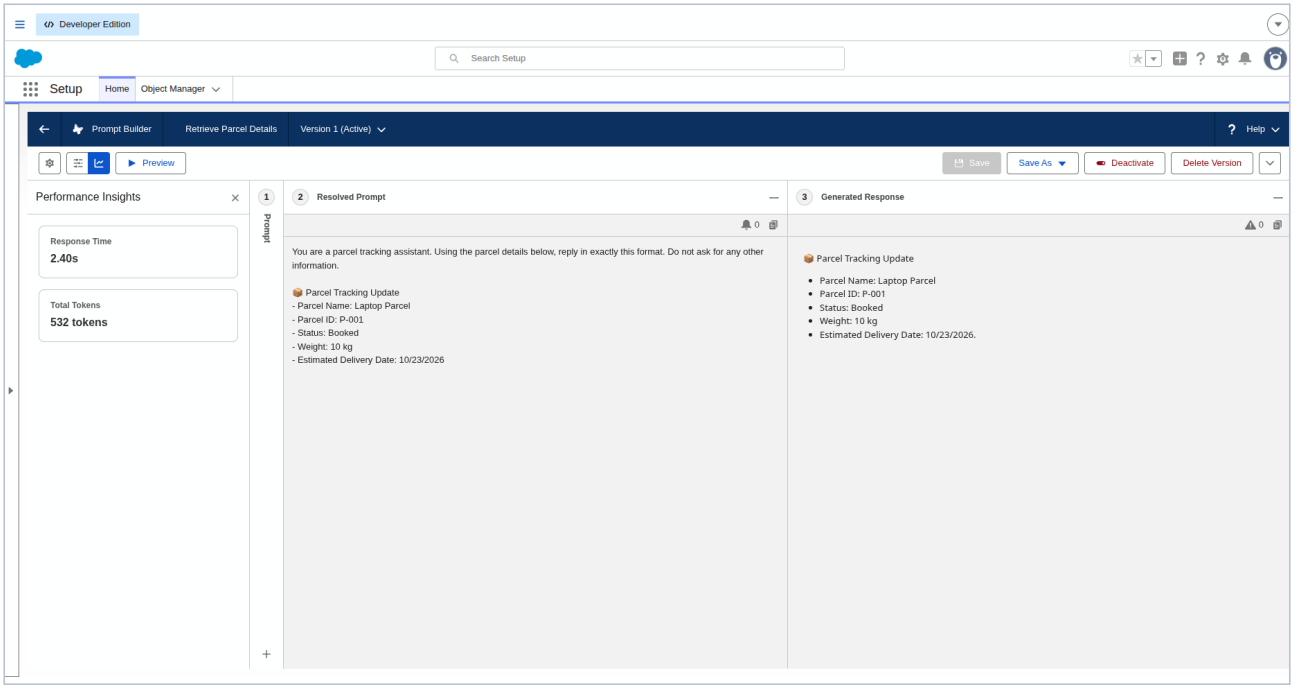


Figure 3: Prompt template preview - resolved prompt and generated response (Version 1, Active)

Milestone 6: Flow

Created an Autolaunched Flow, **Parcel Details**, with an input variable *ids* and an output variable *output*. Steps: Get Records (Parcel where Parcel_ID__c equals ids), Action (the prompt template, with the retrieved Parcel as the object to summarize), and Assignment (output set to the Prompt Response). The Flow is active.

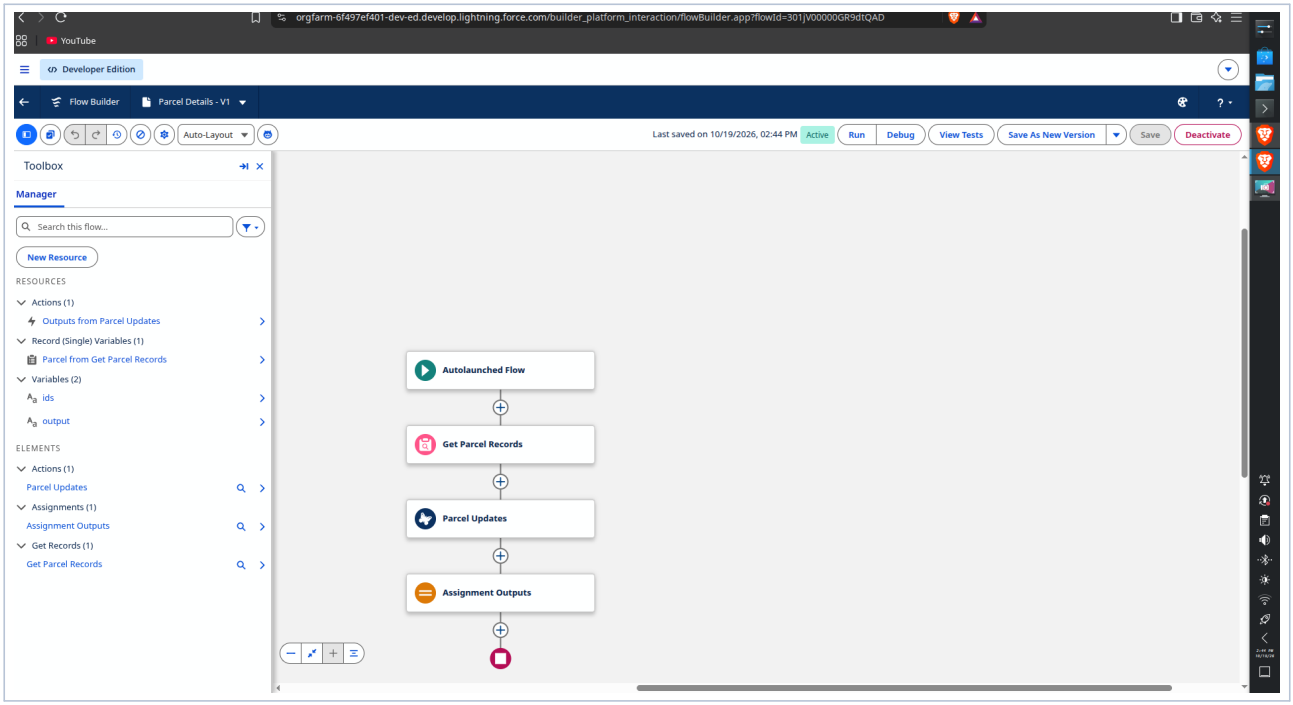


Figure 4: Parcel Details Flow - Get Parcel Records, Parcel Updates, Assignment Outputs (Active)

Phase 3: UI/UX Development and Customization

Milestone 1: Permission set

Created the **Swift Ship** permission set with Read, Create and Edit on Parcel, Delivery, Sender and Receiver, and visible field permissions.

Milestone 2: Assigning permissions

Assigned the permission set to the user who runs the agent. The Einstein Service Agent user was not present in this org, so the permission set was assigned to the admin user and the employee agent runs as that user.

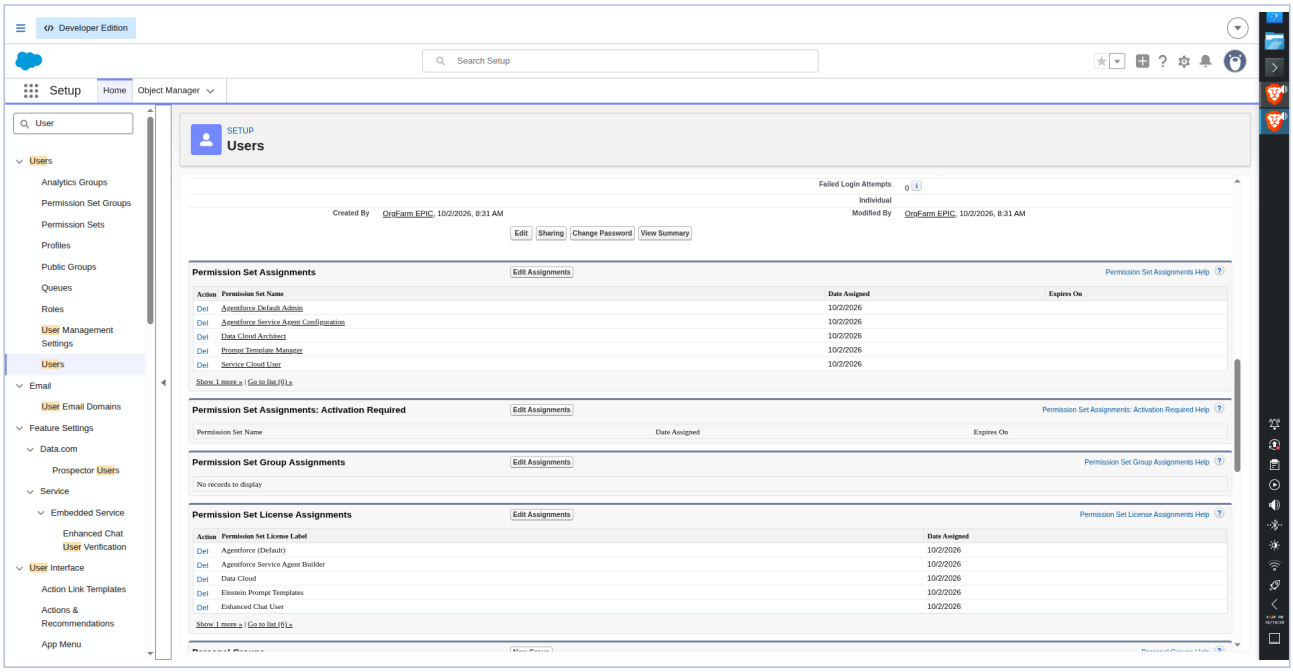


Figure 5: Permission set assignments on the user record

Milestone 3: Org setup for Agentforce

Turned on Einstein, Global Languages, Deploy Prompt Templates, Agentforce and the Agentforce (Default) Agent.

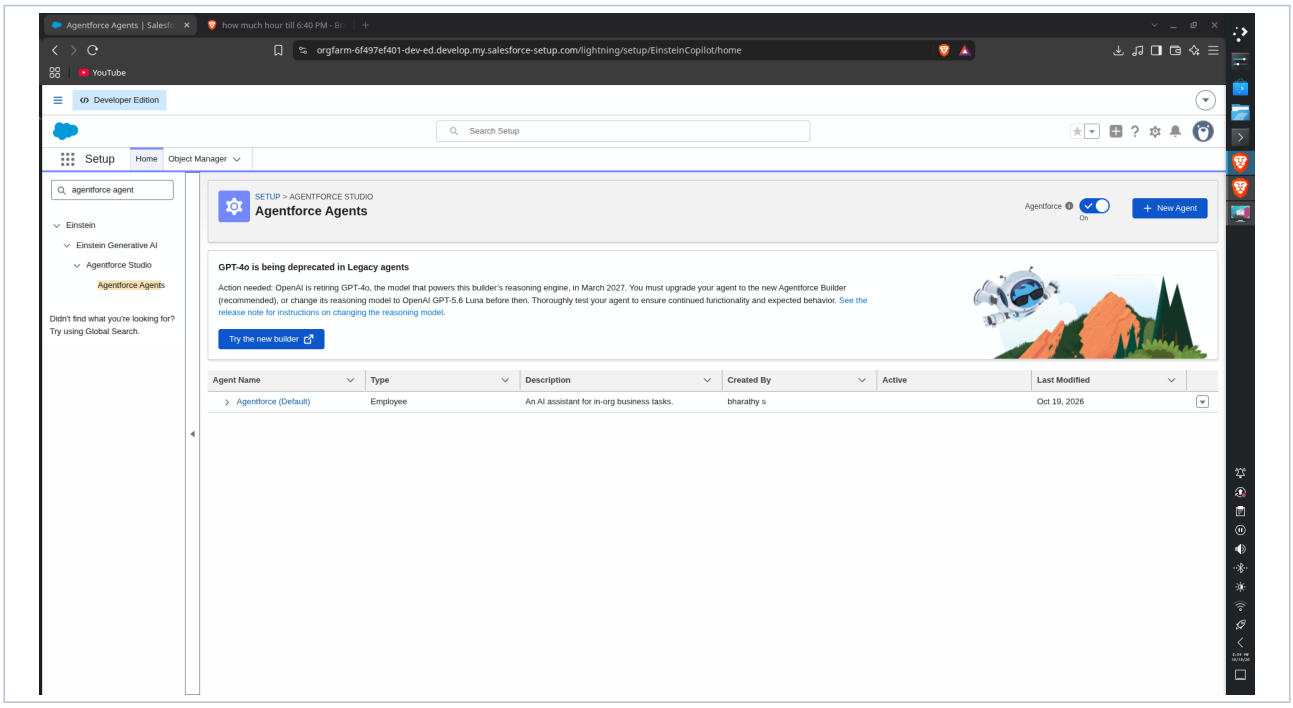


Figure 6: Agentforce enabled - Agentforce (Default) agent listed in Agentforce Studio

Milestone 4: Building the agent

Created the **Parcel Details Inquiry** subagent, which handles requests for parcel details by asking for the Parcel ID. Created an agent action from the Parcel Details Flow, with *ids* as a required input collected from

the user and *output* shown in the conversation. Added the action to the subagent, renamed the agent **SwiftShip Tracker** and activated it.

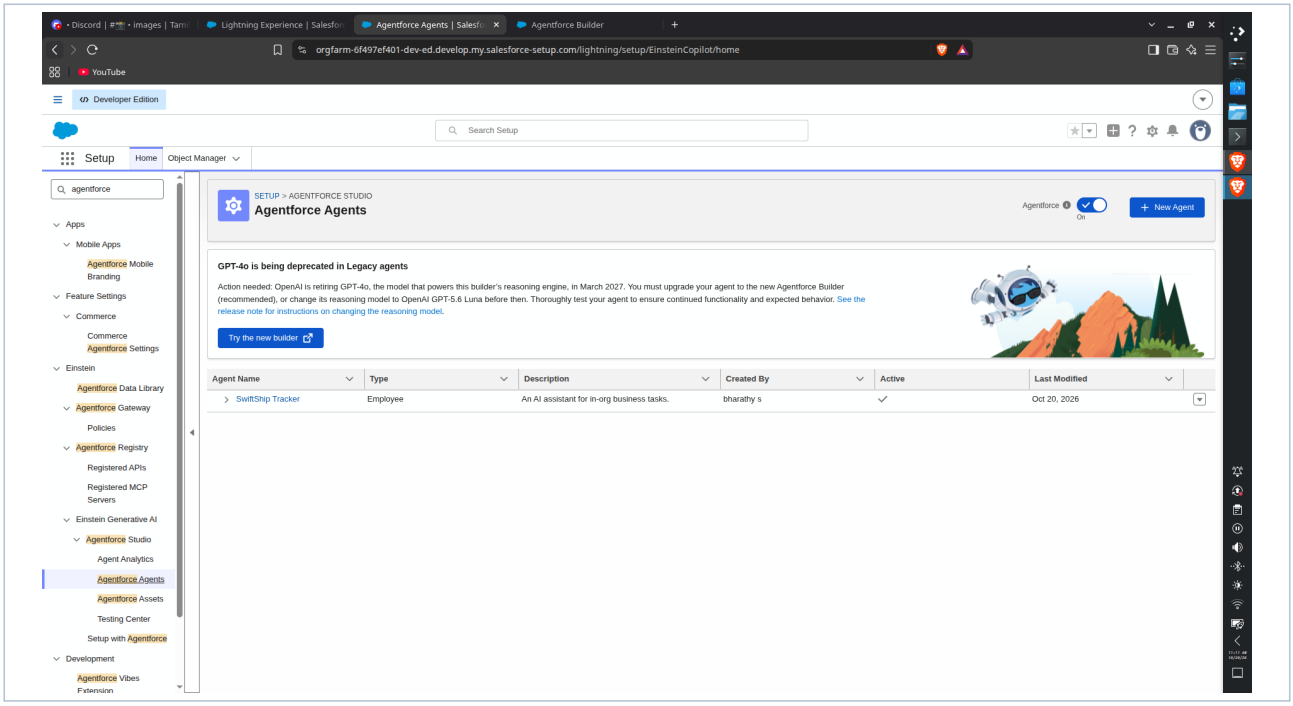


Figure 7: SwiftShip Tracker agent listed and active in Agentforce Agents

Phase 4: Testing and Security

Verification: the SwiftShip Tracker agent is listed and active; the Parcel Details Inquiry subagent is selected for parcel questions; the Parcel Details Flow is active and linked as the subagent's action.

Agent test: the user asked for parcel details, the agent asked for the Parcel ID, and the user replied P-001. The trace shows the Flow action called with ids = P-001, and the agent returned the Parcel Tracking Update for Laptop Parcel (Status Booked, Weight 10 kg).

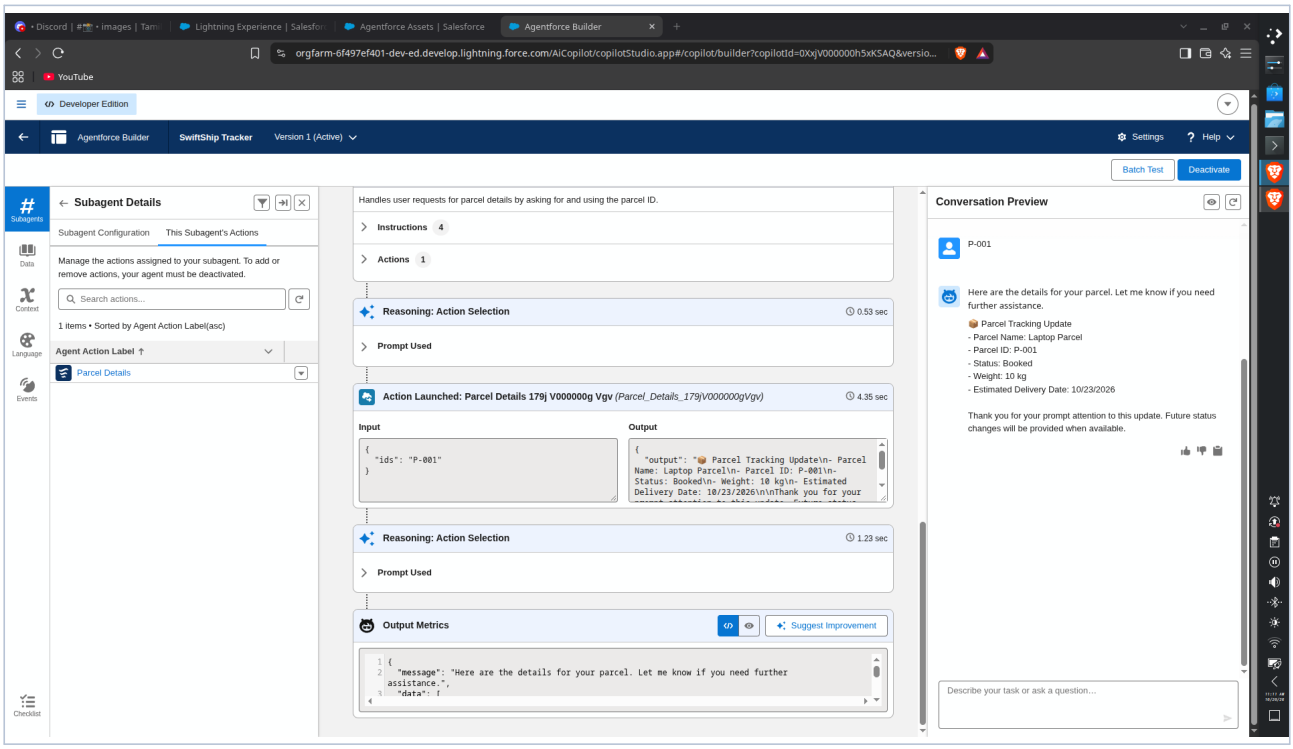


Figure 8: Agent test - conversation preview and reasoning trace showing the Flow action input and output

Phase 5: Deployment, Documentation and Maintenance

The agent was tested in the Developer Org to check topic recognition, Flow execution and data retrieval from custom objects. Production testing (UAT, sandbox, CI/CD) was not performed because Developer Edition orgs are standalone and enterprise deployment is outside the scope of this prototype.

Maintenance and troubleshooting: keep parcel data accurate; monitor for misrouted agent queries and failed Flow runs; update Flow logic, refine subagent descriptions and improve prompts as needs change.

7. Functional and Performance Testing

Test case	Description	Expected result	Actual
TC-01	Create the four custom objects	Objects appear in Object Manager	Pass
TC-02	Create a Parcel record	Parcel ID auto-numbers as P-001	Pass
TC-03	Link Sender, Receiver and Delivery to the Parcel	Lookups save and display	Pass
TC-04	Preview the prompt template with Laptop Parcel	Formatted parcel update is returned	Pass
TC-05	Activate the Flow	Flow shows Active	Pass
TC-06	Ask the agent for parcel details	Agent asks for the Parcel ID	Pass
TC-07	Give the agent P-001	Agent returns name, ID, status, weight and date	Pass
TC-08	Permission set assigned to the user	Assignment is listed on the user	Pass

Performance: in the test trace, subagent and action selection each took about 0.5 seconds and the Flow action took about 4 seconds.

8. Advantages and Disadvantages

Advantages

- Centralized parcel and customer information in Salesforce
- Better visibility into parcel booking and tracking
- Agentforce enables conversational parcel-status assistance without contacting support
- Prompt Builder gives consistent, formatted answers
- Reduces manual effort for customers and support teams
- Can scale with Experience Cloud and external integrations

Disadvantages

- Requires accurate and up-to-date parcel data
- Agent responses depend on well-configured prompts, subagents and actions
- Depends on Salesforce configuration and permissions
- Flows and prompts need maintenance as processes change
- Notifications, reports and dashboards still need to be built

9. Conclusion

SwiftShip Tracker shows how Salesforce can be used to build a centralized parcel management and tracking solution. Custom objects, relationships, a Prompt Builder template, a Flow and an Agentforce agent let a user retrieve shipment information by Parcel ID without depending on manual support. It forms a practical foundation for a scalable parcel management system.

10. Future Scope

- Record-triggered Flows and Email Alerts for automatic status updates and notifications
- Validation rules, Apex triggers and Batch Apex
- Reports and dashboards for parcel status and delivery performance
- Experience Cloud site for customer self-service tracking
- More agent actions, such as updating parcel status
- Integration with external logistics systems
- Sandbox testing and CI/CD deployment

11. Appendix

- **A. Custom objects:** Parcel, Delivery, Sender, Receiver
- **B. Automation:** Autolaunched Flow (Parcel Details)
- **C. AI:** Agentforce agent (SwiftShip Tracker), Prompt Builder template (Retrieve Parcel Details), subagent (Parcel Details Inquiry)
- **D. Security:** Swift Ship permission set